

REPRODUCTIVE SYSTEMS FOR NORTH AMERICAN DAIRY CATTLE HERDS

Ricardo C. Chebel^{1,2}, DVM, MPVM and Eduardo S. Ribeiro³, DVM, MSc, PhD

^{1,2}Associate professor, Department of Large Animal Clinical Sciences, Department of Animal Sciences, University of Florida, Gainesville; 2030 Shealy Dr., Gainesville, FL 32608;

rcchebel@ufl.edu and

³Assistant professor, Department of Animal and Poultry Science, University of Guelph; ²⁵⁰

Stone Rd E, Guelph, ON N1G 2W1, Canada; ribeiro.es@ufl.edu

Corresponding author: Ricardo C. Chebel

The authors have nothing to disclose.

Key words:Dairy cattle, reproductive efficiency

Key points:

- Reproductive inefficiency compromises the health and longevity of individual cows and the profitability of herds.
- In the average dairy herd, the combination of estrus detection and ovulation synchronization protocols yields the best economic return.
- Genomic selection of animals is particularly profitable in situations when little is known about their genetic potential.
- Biosensor systems in milking parlors may allow for the design of reproductive strategies tailored for cows according to their physiological needs while optimizing economic return.

INTRODUCTION

Reproductive management of dairy animals has been marked by tremendous progress in the past 50 years, from the creation of prostaglandin drugs (PGF_{2α}) that allow for the synchronization of estrous in the 70s to the implementation of on-farm *in vitro* embryo production programs and the utilization of genomic selection to aid in breeding strategies. Such progress has been realized because reproductive efficiency has long been identified as critical for the profitability of dairy herds. Herds with efficient reproductive programs enjoy the benefits of having a large proportion of cows in the most productive phase of lactation¹, greater availability of replacement animals, greater genetic progress^{2,3}, reduced proportion of reproduction culls^{4,5}, reduced cost of reproductive programs^{2,4}, and improved health. Furthermore, improved reproductive performance might reduce emission of greenhouse gases because of reduced number of animals per unit of production and reduced retention of replacement animals^{6,7}.

For years, fertility in dairy cows has declined in multiple regions of the world and diverse production systems^{8,9,10,11}. Changes in cow physiology associated with greater milk production, nutritional management, housing, increased herd size, reduced expression of estrus, and genetic makeup of the dairy cattle population have been identified as reasons for the decrease in reproductive efficiency^{8,12,13}. Nonetheless, the downward trend in reproductive efficiency has been somewhat halted because of the tremendous progress in our knowledge of reproductive biology, physiology, and management¹⁴. Calving interval (**CI**), for example, has declined significantly in the past 10 years (Figure 1A) most likely because of reduced calving to first AI interval (Figure 1B and Figure 1C)¹⁴. The positive trend in daughter pregnancy rate (**DPR**) observed in the past 10 years (Figure 1D)¹⁴ may be a result of more controlled voluntary waiting periods (**VWP**) and shorter calving to first AI intervals. It is not possible to discard, however, the

importance of modern genetic selection programs that have focused on milk yield, productive life, reproduction, and health^{3,6,13}.

REPRODUCTIVE EFFICIENCY AND DAIRY HERD PROFITABILITY

The key products of dairy herds are milk and calves (88% and 12%, respectively, of gross income of dairies)¹⁵, whereas the largest expenses are associated with feeding the herd (60% of total operating costs) and rearing replacement animals¹⁵. Reproductive efficiency affects profitability of dairies in several manners that are not always easy to account for. Improved reproductive performance has many beneficial effects: increases efficiency of milk production by shifting the milking herd to a more productive phase of the lactation¹, improving income over feed cost (IOFC) and milk yield per day of CI¹⁶; reduces reproductive culls⁵, reducing the need for replacement animals and increasing the percentage of lactating herd that is $\geq 2^{\text{nd}}$ lactation^{4,15}; improves genetic gain because of more selective culling of lactating cows and more stringent selection of replacement animals^{13,15}; and, reduces the cost of reproductive interventions^{2,4}. Significant improvement in reproductive performance, however, results in a greater proportion of the adult herd dry⁴, demanding proper planning to accommodate dry cows and maternity needs.

The lactation curve and the dry matter intake (DMI) of mature cows determine that the efficiency of milk production is significantly greater during early vs. late lactation (Figure 2A and 2B)¹⁶. Consequently, highest efficiency of milk production is achieved when the majority of the cows in a herd conceive within the first 100 days in milk (DIM). To illustrate this point, let us assume that two herds have a VWP of 50 d, that the average DIM at first AI of 60 d, and that cows are eligible to become pregnant for 13 consecutive cycles (from 50 to 312 DIM). In a herd with average milk yield of 12,500 kg/305-dof lactation, improving 21-d pregnancy rates (21-d

PR; percentage of eligible cows that become pregnant every 21 d) from 12 to 33% would reduce the average CI by 63 days (from 440 to 377) and result in an increase of 7% in the IOFC and 1.51 kg of milk/cow per day of CI (551 kg of milk/cow per year; Figure 3A)¹⁶. If the same improvements in 21-d PR are achieved in a herd with average milk yield of 9,000 kg/305-d of lactation, IOFC would increase 8% and milk yield would increase by 1.11 kg/cow per day of CI (405 kg of milk/cow per year; Figure 3B)¹⁶. The economic gains of improving reproductive performance in herds with higher productivity and more persistent lactation curve may not be as dramatic¹⁷.

The economic value of improved reproductive performance is not easily determined, is very particular for given production systems, and is affected by several factors. Several researchers have created models to estimate the value of increasing 21-d PR or reducing the calving to pregnancy interval (days open) according to different scenarios. De Vries suggested that the cost of a day open varies from \$0 to \$6.00 depending on DIM, milk production and price, feeding costs, persistency of lactation curve, lactation number, culling policies and availability of replacement heifers¹⁷. Groenendaal et al. estimated the cost of a day open to be \$1.25, \$2.10 and \$2.75 up to 90, 150 and 210 DIM, respectively¹⁸. Researchers have also created models to estimate the value of newly established pregnancies and the cost of pregnancy losses. De Vries proposed that the difference in retention pay off (**RPO**; economic benefit of keeping a cow in the herd versus culling and replacing it with a replacement heifer) of pregnant and non-pregnant cows provides an estimate of the value of a new pregnancy (assuming equal milk yield, DIM, lactation curve persistency, and lactation number)¹⁹. According to this theory, the value of a pregnancy is less early postpartum because there is little difference in RPO between pregnant and non-pregnant cows. The value of a pregnancy increases dramatically in later stages

of lactation. For example, when evaluating two cows in later stages of lactation with equal milk yield, lactation curve persistency, and lactation number, it would be more profitable to replace the non-pregnant cow for a replacement heifer, whereas it is profitable to keep the pregnant cow¹⁹. According to De Vries, establishing a new pregnancy in the early stages of lactation is highly valuable for low-producing cows and cows with low persistency of the lactation curve but establishing a new pregnancy too early postpartum in high-producing cows and cows with high persistency of the lactation curve may result in economic losses¹⁹. The value of a new pregnancy has been estimated to range from \$-100 to \$500¹⁹ and from \$128 to \$232²⁰. Cost of a pregnancy loss (\$0 to \$1,373¹⁹; \$128 to \$897²⁰), on the other hand, increases as stage of lactation, stage of gestation and milk yield increase¹⁹.

The aforementioned research may lead to the suggestion that reproductive management should be tailored to the individual to optimize economic return. Although this may be possible in small herds, and in the future in large herds, today's reality of large herds demands reproductive management systems that are robust and meet the needs of the majority of cows in a group. With that in mind, many herds adopt robust reproductive strategies for first postpartum AI and re-insemination of all cows²¹, and some herds adopt different VWP according to parity. The average VWP is approximately 55 d in US dairy herds, with nearly 75% of dairies having a VWP of 41 to 60 DIM (Figure 4)²², and the average interval from calving to first AI is 78 ± 0 and 74 ± 0.1 d for Holstein and Jersey cows, respectively¹⁴. The VWP should be established based on the anticipated optimum time for establishment of pregnancy and on the reproductive efficiency of the herd. In most herds it is not possible to precisely control when a cow becomes pregnant and it is necessary to adopt a relatively short VWP. Figure 5 depicts the recommended VWP for herds according to the desired interval from calving to pregnancy and 21-d PR,

maintaining a maximum of 13 reproductive cycles. To achieve an average days open of 110 d, herds with 21-d PR < 25% would have to have VWP < 50 DIM, which is often associated with reduced fertility. On the other hand, herds that desire to achieve 150 d open and have 21-d PR >25% may have VWP > 80 DIM. De Vries suggested that the ideal interval from calving to pregnancy is, on average, 133 d for 1st lactation cows and 112 d for 2nd lactation cows, but could be reduced to 63 and 51 d for cows with low production and low lactation persistency¹⁷. In summary, herds with reduced milk yield, poor lactation curve persistency and low 21-d PR must have short VWP and aggressive reproductive management. On the other hand, herds with persistent lactation curves, high milk yield, and elevated 21-d PR may benefit from extending the VWP.

Greater reproductive efficiency reduces the number of reproductive culls, which affects replacement policies because dairymen may make culling decisions based on economic aspects rather than biological considerations¹⁸. In most scenarios, reduced reproductive efficiency forces managers to either reduce selection pressure, otherwise number of lactating cows may be reduced affecting cash flow, or purchase replacement heifers to replace cows culled for reproduction failure, jeopardizing genetic improvement and biosecurity of the herd. Dairies that enjoy efficient reproductive management often have a surplus of replacement heifers⁴. Rearing replacement heifers accounts for approximately 25% of the total cost of dairy operations¹⁵, but herds with good reproductive performance may be more selective in replacement selection and may profit from the sale of heifers for dairy purposes.

REPRODUCTIVE EFFICIENCY AND BREEDING STRATEGIES

Development of PGF_{2α} in the 70s produced significant advancements in reproductive management of dairy cattle. Although PGF_{2α} treatment produces luteolysis and induces estrus in a large proportion of cyclic animals, accuracy and efficiency of estrous detection are variable and dependent on animal, environmental, and management factors^{23,24,25}. In the late 90s, ovulation synchronization protocols (OVSP; e.g. Ovsynch²⁶) were developed, allowing for the fixed time AI of cows while producing acceptable pregnancy per AI (**P/AI; number of cows pregnant divided by the number of cows inseminated**). Implementation of OVSP for timed AI increases yearly profit per cow by \$30 compared with estrous detection alone²⁷ and reduces the cost per pregnancy by \$80 to \$148 compared with AI on detected estrus (**DE**) alone¹⁷.

The use of OVSP for timed AI of lactating dairy cows has become popular to the point of some producers using it exclusively to manage their herds. Questions arise, however, to what is the most profitable reproductive management strategy (e.g. 100% OVSP, 100% AI on DE, combination of OVSP and AI on DE). To answer this question, it is important to consider a few facts. One of the most important is the fact that identification of non-pregnant cows following AI may be done by observed return to estrus or by non-pregnancy diagnosis. In most herds that adopt estrous detection, 40 to 60% of non-pregnant cows are observed in estrus and re-inseminated between 3 and 28 d after previous AI (average 22 d). On the other hand, non-pregnancy diagnosis by transrectal ultrasonography may be done as early as 26 d after AI with proper accuracy and efficiency and, if an aggressive OVSP for timed AI is adopted, cows could be re-inseminated as early as 29 d after previous AI. Thus, the interval to re-insemination in herds that adopt estrous detection may be 4-5 d shorter, if not shorter, compared with herds that do not re-inseminate cows based on signs of estrus. Although this may lead to the suggestion that adopting estrous detection is advisable, this is not always the case. In herds in which efficiency

of estrous detection is high but accuracy is low, it may be deleterious to increase the number of cows inseminated in estrus. Giordano et al. indicated that when P/AI of cows inseminated in estrus is 25%, increasing the percentage of cows inseminated in estrus from 30 to 80% would decrease net return per cow per year from -\$7.91 to -\$14.84/cow per year (compared with 100% OVSP and timed AI)²⁸. Tenhagen et al. demonstrated that when P/AI of cows inseminated in estrus is adequate (35 to 40%) and estrous detection rate is approximately 55%, herds may rely more heavily on AI on DE but when estrous detection rate is extremely low (< 30%) it may be more advantageous to implement aggressive OVSP for timed AI²⁹. Galvão et al. simulated in a herd of 1,000 cows the best economic outcomes of reproductive programs that employed 100% OVSP and timed AI with differing levels of compliance (85 and 95%), 100% AI on DE with differing levels of efficiency (40 and 60%) and accuracy (85 and 95%), and a combination of OVSP and timed AI and AI on DE⁴. Pregnancies per AI of the different protocols were estimated from the literature. According to this simulation, the reproductive programs that resulted in the best return per cow per year, the greatest 21-d PR, and the shortest average DIM were programs that combined OVSP for timed AI and AI on DE⁴. As expected, poor compliance to OVSP and poor estrus detection efficiency and accuracy had important detrimental effects to the profitability of the herds (Figure 6)⁴. According to Kalantari and Cabrerathe best economic return was obtained when 80% of cows were inseminated on DE and P/AI of cows inseminated on DE was 35% and P/AI of cows inseminated at fixed time following OVSP were 30 and 28% for first AI and re-inseminations, respectively³⁰. This raises a very important topic that is often overlooked when discussing reproductive management of lactating dairy cows and heifers: labor force. Proper training and education of the labor involved in all aspects of reproductive management is fundamental for the success of such programs. Veterinarians and consultants

have a prime opportunity to make themselves indispensable to dairy producers when they become involved in training, educating, and monitoring the performance of the labor involved in these tasks. In conclusion, no one reproductive management strategy fits all dairy herds in the US. Before a reproductive management strategy is recommended, it is important to evaluate facilities (e.g. freestall, open lots, tie stalls) and management (e.g. severe overstocking), personnel (e.g. training, education), herd composition (e.g. breed, percentage of first lactation cows), adoption of new technologies, record keeping, and cow health.

Automated estrous detection systems (e.g. pedometers or activity monitors) have had greater penetration in the US dairy industry in the past 10 years. These systems predict estrus with 70 to 80% accuracy based on changes in behavioral patterns³¹ and their performance is comparable to that of mount detection devices³². Although many have suggested that the adoption of automated estrus detection systems in the management of dairy herds may eliminate the need for OVSP for timed AI and reduce hormone use in dairy herds, this is not necessarily true. Unquestionably, herds with poor estrous detection efficiency and accuracy may benefit significantly from the use of automated estrous detection systems. The automated estrous detection systems, however, will not and should not detect in estrus cows that do not display estrus (e.g. anovular cows). In recent experiments, 7 to 44% of cows were not detected in estrus by such systems^{32,33,34}. Therefore, most herds that adopt automated estrous detection systems still need to implement OVSP for timed AI of cows that do not display estrus. The cost of the automated estrus detection systems may easily be compensated by significant improvements in estrous detection efficiency and accuracy, which would lead to improved 21-d PR. Thus, one may assume that herds with low estrous detection efficiency and poor P/AI of cows inseminated on DE would realize economic gains when adopting automated estrous detection systems,

particularly when such systems have long durability. Furthermore, automated estrous detection systems may help to optimize the timing of insemination of heifers and lactating cows with sex-sorted semen and improve their P/AI^{35,36}, which may be an attractive strategy for herds that desire a larger pool of replacement heifers.

According to the 2009 NAHMS, 25% of the dairy calves born in the US are from cows sired by natural service²². Although many producers defend the use of natural service with the belief that reproductive performance may be improved compared with AI systems, epidemiological studies have demonstrated that reproductive performance is not altered³⁷ or worsened³⁸ for natural service compared with AI based on DE or OVSP systems. Lima et al. compared reproductive performance and economic return of reproductive management based on 100% OVSP for timed AI versus 100% natural service^{39,40}. In this experiment, the OVSP for timed AI was very aggressive to assure timely re-insemination of cows that failed to conceive to a previous AI. Similarly, sires were aggressively managed and underwent a breeding soundness exam every 3 months (including trichomoniasis screening), the bull to open cow ratio was 1:20, exposure period to rest period pattern was 14:14d, and bulls were evaluated weekly for health problems and replaced if necessary. In this experiment, the median days to pregnancy was slightly shorter for the natural service (111 vs. 116 d) but, because of the cost of feeding sires and gains in genetic merit by using semen **for timed AI**, the economic return was approximately \$32.47/cow per year greater for the 100% OVSP for timed AI versus the 100% natural service^{39,40}. In summary, natural service may be a reproductive strategy for managers that do not have the ability to manage the systems necessary for AI (estrous detection or OVSP, labor, semen inventory, **inventory of** AI consumables); however, natural service is not a strategy to improve reproductive performance or genetic composition of the herd.

REPRODUCTIVE EFFICIENCY AND HEALTH

Although constantly overlooked, reproductive efficiency also is associated with overall health of dairy herds and individual cow health. As aforementioned, improved reproductive efficiency reduces both the number of reproductive culls⁵ and the need for replacement heifers⁴, while the availability of replacement heifers increases⁴. Therefore, herds with efficient reproductive management that are not expanding do not need to purchase replacement heifers to maintain a constant number of lactating cows in the herd, reducing the risk of a breach in biosecurity and introduction of contagious and infectious diseases.

Improved reproductive performance is also expected to improve the health of individual cows. Success of the reproductive program and gestation length determine the CI. In most herds the dry period is constant (45 to 60 d in most US dairy herds)²² and extending the CI results in cows spending a larger amount of time in the less efficient stages of lactation¹. Therefore, cows that do not conceive promptly after calving tend to gain a significant amount of body fat reserves during the final stages of lactation, even when herds adjust rations according to milk yield. According to a large data set (8,989 lactations from two CA herds), BCS at dry-off was mainly explained by 305-d mature equivalent milk yield, interval from calving to establishment of pregnancy, and number of lactations (Figure 7; Chebel and Mendonça, unpublished data). In general, there was a positive correlation between interval from calving to conception and BCS at dry-off and a negative correlation between 305-d mature equivalent milk yield and BCS at dry-off.

Excessive accumulation of fat poses a significant risk for the health of periparturient dairy cows because obese cows have more dramatic immunosuppression during this period than

thinner cows^{41,42}. Predisposing factors for reduced innate immune function during the prepartum period are hormonal alterations associated with parturition⁴³, reduced DMI⁴⁴, increased concentrations of NEFA and BHBA^{44,45,46,47}, and decreased concentration of IGF-1^{48,49}. Consequently, conditions that limit DMI and exacerbate negative energy balance during the periparturient period increase the risk of infectious diseases⁵⁰. Obese cows (BCS ~ 4.0) had reduced DMI during the last 21 d of gestation and had a more dramatic decrease in DMI from 21 to 1 d before calving⁵¹. Furthermore, BCS 21 d before calving explained 9.7% of the variability in DMI from 21 to 1 d before calving⁵¹. Consequently, obese cows are more likely to lose BCS and body weight during the periparturient period than thinner cows^{52,53}. In a model that explained 38.7% of the variability in BCS change during the dry period, BCS at dry-off explained 90.6% of its variability (Figure 8; Chebel and Mendonça, unpublished data). The association between BCS loss during the dry period and incidence of postpartum diseases also was investigated. Loss of BCS during the prepartum period increased the likelihood of diseases such as retained fetal membranes, metritis, and digestive disorders and increased the likelihood of cows being removed from the herd within 30 and 60 DIM (Table 1; Chebel and Mendonça, unpublished).

In summary, excessive BCS at dry-off reflects poor milk yield and/or poor reproductive efficiency. Obese cows have compromised immune and metabolic functions in the prepartum period and are more likely to present health disorders in the postpartum. Because occurrence of metabolic and infectious diseases early postpartum has a negative impact on reproductive performance^{25,54}, a revolving door scenario is established jeopardizing the life time production of cows.

THE ROLE OF GENOMIC SELECTION IN REPRODUCTIVE MANAGEMENT OF DAIRY HERDS

Genetic selection of the Holstein breed has been focused mainly on milk yield until approximately 10 years ago. Unfortunately, a negative correlation exists between genetic traits for milk yield and other traits, including reproductive efficiency^{55,56}. Consequently, compared with genetically inferior cows for milk yield, a population of genetically superior cows for milk yield presents important physiological and behavioral alterations that may, at least in part, be the cause of reduced reproductive efficiency observed from the 50s to the beginning of the current century. A breeding project that started in 1964 at the University of Minnesota provides evidence that genetic selection for high milk yield is associated with physiological alterations of dairy cows that affect fertility. Cows bred continuously to sires and sons of sires with PTA-milk average for the 60s (unselected = $6,890 \pm 403$ kg/305-d lactation) have milk yield $\sim 4,000$ kg/305-d lactation smaller than cows bred continuously to sires with the highest PTA-milk from each year (selected = $11,078 \pm 329$ kg/305-d lactation)^{57,58}. Unselected and selected cows had similar body weight, degree of fat depots mobilization during the peripartum period, and energy balance during the postpartum period^{59,60}. Unselected cows, however, had lower growth hormone concentration during the lactation and higher insulin like growth factor-1 concentration until 12 weeks of lactation^{59,60,61}. These differences in somatotrophic axis are critical for resumption of ovarian cycles following parturition⁶². Consequently, unselected cows resumed ovarian cycles ~ 10 d sooner and had greater progesterone concentration during the first three estrous cycles postpartum compared with selected cows^{60,63}. These experiments demonstrate that genetic selection for high milk yield alters reproductive physiology of lactating dairy cows directly or indirectly (as suggested by Sangsritavong et al.⁶⁴).

Conventional methods of genetic selection [parental average (**PA**) and predicted transmitting ability (**PTA**)] are somewhat limited in their ability to improve genetic progress of traditional fertility traits (e.g. DPR, CI) because such traits have low heritability^{3,65,66,67}. Heritability of productive traits range from 0.30 (mature equivalent milk, fat, and protein yield) to 0.58 (fat percentage), whereas heritability of reproductive traits range from 0.04 (interval from calving to first insemination) to 0.23 (interval from calving to first ovulation). The low heritability of **traditional** reproductive traits may reflect their dependence on a multitude of cow and managerial factors. For example, DPR is the speed at which animals become pregnant assuming a 60-day VWP that is highly affected by the actual VWP adopted on-farm, use of fixed time AI, and use of natural service. On the other hand, CI is dependent on the interval from calving to resumption of ovarian cycles, estrous expression and estrous detection efficiency, establishment and maintenance of pregnancies, and length of gestation⁶⁸. Despite the low heritability of traditional reproductive traits, the implementation of genetic evaluation for DPR starting in 2003 is believed to have contributed to increases in reproductive performance of Holstein and Jersey breeds³. Physiological characteristics of cows that are less dependent on management and affect reproductive performance (**e.g. postpartum resumption of ovarian cycles and pregnancy loss**) may present higher heritability and may be better candidates for genetic selection^{12,69,70}.

The generation of PA and PTA data demands large investments of time and resources, which still do not guarantee 100% accurate information because of inherent confounding of some phenotypes (**e.g. DPR and CI**)⁷¹. In recent years, cattle breeders and producers have gained access to somewhat affordable genomic evaluation. Initially (late 2007), only young males and elite females were tested using the 50,000 (**50K**) bovine single nucleotide polymorphisms (**SNP**)

probe because cost limited its broader use in the cattle population. Since 2010, however, a more affordable array (**3K** probe) has been available to the dairy industry. The low-density array (3K) may be used effectively for imputation of the 50K probes and produces results (genomic-predicted transmitting ability; **GPTA**) almost as accurate as the results produced with the 50K probes^{72,73}. The use of GTPA for selection of future sires presents economic advantages because of its higher reliability compared with PA or PTA, because it expedites genetic gains through shortening of generation intervals, and because it allows for the elimination of lower GPTA animals soon after birth⁷⁴. In recent years, 90% of young Holstein and Jersey sires have been selected using GPTA⁷⁵. Questions remain, however, on how commercial producers may use GPTA information in their reproductive programs. The cost of genomic testing is ~\$40 per animal, demanding that a clear strategy to use the GPTA information generated be designed to result in profits. Weigel et al. simulated lifetime net merit (LNM\$) gains that dairy herds may realize by using GPTA for selection of females taking into consideration reliability of the test and selection strategy⁷⁶. According to this simulation, reliability of selection would be smallest when heifer calves of unknown pedigree are selected based on phenotype alone and reliability of selection would be maximum when 6th lactation cows with complete pedigree are selected based on GPTA. According to the authors, genomic testing is more valuable when less is known about an animal's genetic potential (heifer calves or animals with no pedigree)⁷⁶. In fact, use of GPTA for selection of heifer calves with no pedigree resulted in increase in LNM\$ of \$66 to \$259 per animal, whereas use of GPTA for selection of heifer calves with full pedigree resulted in increase in LNM\$ of \$19 to \$87 per animal⁷⁶. Claus et al. also estimated the benefits of using GPTA to select replacement heifers according to different scenarios⁷⁷. In general, an economic advantage was observed when using GPTA to select replacement heifers, particularly when: little was

known about their genetic potential, replacement heifers were abundant, cost of testing was low, accuracy of GPTA was significantly higher than PTA, and benefits of genetic improvement were higher⁷⁷. In summary, genomic testing may result in significant economic gains when a clear strategy exists to make use of the GPTA data. Increase use of GPTA for selection of replacement heifers will improve the quality of genetic information of herds, increasing the reliability of PTA selection and, possibly, reducing the economic advantage of using GPTA, unless niche markets are explored (stock farms) or cost of genomic testing is reduced significantly.

FUTURE CONSIDERATIONS

Biosensor systems within the milking parlor (e.g. Afilab system and Herd Navigator) are currently available in Israel, Europe, Canada, and a few select markets of the US. Such systems allow the daily monitoring of dairy cows for several metabolites and biological markers that are used to estimate energy balance, identify occurrence of metabolic (e.g. ketosis) and infectious (e.g. mastitis) diseases, and to predict stage of the estrus cycle and pregnancy status (e.g. in-line determination of progesterone and pregnancy associated glycoprotein concentrations). Such systems coupled with technologies currently available (e.g. GPTA, activity monitors/pedometers, rumination monitors, ultrasonography, OVSP for timed AI, sex-sorted semen) will facilitate the designing of reproductive strategies for specific groups of cows according to their physiological needs while maximizing profitability.

REFERENCES

1. Ferguson JD, Galligan DT. Veterinary reproductive programs. 1999. In Proceedings of 32nd Annual Convention of the American Association of Bovine Practitioners (AABP), Nashville, TN pp. 133-137. American Association of Bovine Practitioners, Opelika, AL, US.
2. Giordano JO, Kalantari AS, Fricke PM, et al. A daily herd Markov-chain model to study the reproductive and economic impact of reproductive programs combining timed artificial insemination and estrus detection. *J Dairy Sci.* 2012; 95(9):5442-60.
3. Norman HD, Wright JR, Hubbard SM, et al. Reproductive status of Holstein and Jersey cows in the United States. *J. Dairy Sci.* 2009; 92:3517-28.
4. Galvão KN, Federico P, De Vries A, Schuenemann GM. Economic comparison of reproductive programs for dairy herds using estrus detection, timed artificial insemination, or a combination. *J. Dairy Sci.* 2013; 96: 2681-93.
5. Pinedo PJ, de Vries A, Webb DW. Dynamics of culling risk with disposal codes reported by Dairy Herd Improvement dairy herds. *J. Dairy Sci.* 2010; 93:2250–61.
6. Bell MJ, Wall E, Russell G, et al. The effect of improving cow productivity, fertility, and longevity on the global warming potential of dairy systems. *J Dairy Sci.* 2011; 94:3662-78.
7. Garnsworthy PC. The environmental impact of fertility in dairy cows: a modeling approach to predict methane and ammonia emissions. *Anim. Feed Sci Tech*, 2004; 112: 211-23.
8. Lucy MC. Reproductive loss in high-producing dairy cattle: where will it end? *J. Dairy Sci.* 2001; 84:1277-93.
9. VanRaden PM, Sanders AH, Tooker ME, et al. Development of a national genetic evaluation for cow fertility. *J. Dairy Sci.* 2004; 87:2285-92.
10. Hare E, Norman HD, Wright JR. Trends in calving ages and calving intervals for dairy cattle breeds in the United States. *J. Dairy Sci.* 2006; 89:365-70.

11. Walsh SW, Williams EJ, Evans ACO. A review of the causes of poor fertility in high milk producing dairy cows. *Anim. Repro. Sci.* 2011; 123:127-38.
12. Royal MD, Flint AP, Woolliams JA. Genetic and phenotypic relationships among endocrine and traditional fertility traits and production traits in Holstein-Friesian dairy cows. *J. Dairy Sci.* 2002; 85:958-67.
13. Weigel KA. Prospects for improving reproductive performance through genetic selection. *Anim. Reprod. Sci.* 2006; 96:323-30.
14. Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013
(<https://www.cdcb.us/publish/dhi/current/reproall.html>)
15. Santos JEP, Bisinotto RS, Ribeiro ES, et al. Applying nutrition and physiology to improve reproduction in dairy cattle. *Soc Reprod Fertil Suppl*, 2010; 67:387-403.
16. Ribeiro ES, Galvão KN, Thatcher WW, et al. Economic aspects of applying reproductive technologies to dairy herds. *Anim Reprod* 2012; 9:370-87
17. De Vries A. Economics of reproductive performance. In: Risco CA, Melendez P (Ed.). *Dairy Production Medicine*. 2011. Hoboken, NJ: Wiley-Blackwell. pp. 139-151.
18. Groenendaal H, Galligan DT, Mulder HA. An economic spreadsheet model to determine optimal breeding and replacement decisions for dairy cattle. *J Dairy Sci.* 2004; 87:2146-57.
19. De Vries A. Economic value of pregnancy in dairy cattle. *J Dairy Sci.*, 2006; 89:3876-3885.
20. Cabrera VE. Economics of fertility in high-yielding dairy cows on confined TMR systems. *Animal*. 2014; 8 (Suppl. 1):211-21.

21. Caraviello DZ, Weigel KA, Fricke PM, et al. Survey of management practices on reproductive performance of dairy cattle on large US commercial farms. *J Dairy Sci.*, 2006; 89:4723-35.
22. National Animal Health Monitoring System (NAHMS). 2009. Dairy 2007. Part IV: Reference of Dairy Cattle Health and Management Practices in the United States, 2007. #N494.0209. Fort Collins, CO: Centers for Epidemiology and Animal Health/USDA/APHIS/VS
23. Vailes LD, Britt JH. Influence of footing surface on mounting and other sexual behaviors of estrual Holstein cows. *J Anim. Sci.*, 1990; 68:2333-9.
24. Wiltbank M, Lopez H, Sartori R, et al. Changes in reproductive physiology of lactating dairy cows due to elevated steroid metabolism. *Theriogenology*, 2006; 65:17-29.
25. Santos JE, Rutigliano HM, Sá Filho MF. Risk factors for resumption of postpartum estrous cycles and embryonic survival in lactating dairy cows. *Anim. Reprod. Sci.* 2009 Feb;110:207-21.
26. Pursley JR, Mee MO, Wiltbank MC. Synchronization of ovulation in dairy cows using PGF2alpha and GnRH. *Theriogenology*. 1995; 44:915-23.
27. LeBlanc S. Economics of improving reproductive performance in dairy herds. *WCDS Adv. Dairy Technol.*, 2007; 19:201-214.
28. Giordano JO, Fricke PM, Wiltbank MC, et al. An economic decision-making support system for selection of reproductive management programs on dairy farms. *J Dairy Sci.*, 2011; 94:6216-6232.
29. Tenhagen BA, Drillich M, Surholt R, et al. Comparison of timed AI after synchronized ovulation to AI at estrus: Reproductive and economic considerations. *J Dairy Sci*, 2004; 87:85-94.

30. Kalantari AS, Cabrera VE. The effect of reproductive performance on the dairy cattle herd value assessed by integrating a daily dynamic programming model with a daily Markov chain model. *J Dairy Sci.* 2012; 95:6160-70.
31. Roelofs J, López-Gatius F, Hunter RH, et al. When is a cow in estrus? Clinical and practical aspects. *Theriogenology.* 2010; 74:327-44.
32. Valenza A, Giordano JO, Lopes G Jr, et al. Assessment of an accelerometer system for detection of estrus and treatment with gonadotropin-releasing hormone at the time of insemination in lactating dairy cows. *J Dairy Sci.* 2012; 95:7115-27.
33. Fricke PM, Giordano JO, Valenza A, et al. Reproductive performance of lactating dairy cows managed for first service using timed artificial insemination with or without detection of estrus using an activity-monitoring system. *J Dairy Sci.* 2014; 97:2771-81.
34. Stevenson JS, Hill SL, Nebel RL, et al. Ovulation timing and conception risk after automated activity monitoring in lactating dairy cows. *J Dairy Sci.* 2014; 97:4296-308.
35. Bombardelli GD, Soares HF, Chebel RC. Time of insemination relative to reaching activity threshold is associated with pregnancy risk when using sex-sorted semen for lactating Jersey cows. *Theriogenology* 2015; doi: <http://dx.doi.org/10.1016/j.theriogenology.2015.09.042>.
36. Sá Filho MF, Mendanha MF, Sala RV, et al. Use of sex-sorted sperm in lactating dairy cows upon estrus detection or following timed artificial insemination. *Anim Reprod Sci.* 2013; 143:19-23.
37. De Vries A, Steenholdt C, Risco CA. Pregnancy rates and milk production in natural service and artificially inseminated dairy herds in Florida and Georgia. *J Dairy Sci* 2005; 88:948-956.

38. Overton MW, Sisco WM. Comparison of reproductive performance by artificial insemination versus natural service sires in California dairies. *Theriogenology*, 2005; 64:603-13.
39. Lima FS, Risco CA, Thatcher MJ, et al. Comparison of reproductive performance in lactating dairy cows bred by natural service or timed artificial insemination. *J Dairy Sci*, 2009; 92:5456-66.
40. Lima FS, De Vries A, Risco CA, et al. Economic comparison of natural service and timed artificial insemination breeding programs in dairy cattle. *J Dairy Sci*, 2010; 93:4404-13.
41. Samartín S, Chandra RK. Obesity, overnutrition and the immune system. *Nutr. Res.* 2001; 21:243–62.
42. Sauerwein H, Bendixen E, Restelli L, et al. The adipose tissue in farm animals: a proteomic approach. *Curr. Protein Pept. Sci.* 2014; 15:146–55.
43. Kimura K, Goff JP, Kehrli ME Jr. Effects of the presence of the mammary gland on expression of neutrophil adhesion molecules and myeloperoxidase activity in periparturient dairy cows. *J Dairy Sci.* 1999; 82:2385-92.
44. Hammon DS, Evjen IM, Dhiman TR, et al. Neutrophil function and energy status in Holstein cows with uterine health disorders. *Vet. Immunol. Immunopathol.* 2006;113:21-9.
45. Hill AW, Reid IM, Collins RA. Influence of liver fat on experimental *Escherichia coli* mastitis in periparturient cows. *Vet. Rec.* 1985; 117:549-51.
46. Kaneene JB, Miller R, Herdt TH, et al. The association of serum nonesterified fatty acids and cholesterol, management and feeding practices with peripartum disease in dairy cows. *Prev. Vet. Med.* 1997; 31:59-72.

47. Rukkamsuk T, Kruij TA, Wensing T. Relationship between overfeeding and over conditioning in the dry period and the problems of high producing dairy cows during the postparturient period. *Vet. Quart.* 1999; 21:71-7.
48. Heemskerk VH, Daemen MARC, Buurman WA. Insulin-like growth factor-1 (IGF-1) and growth hormone (GH) in immunity and inflammation. *Cytokine Growth Factor Rev.* 1999; 10:5-14.
49. Kasimanickam RK, Kasimanickam VR, Olsen JR, et al. Associations among serum pro- and anti-inflammatory cytokines, metabolic mediators, body condition, and uterine disease in postpartum dairy cows. *Reprod. Biol. and Endocrinol.* 2013; 11:103.
50. Huzzey JM, Veira DM, Weary DM, et al. Prepartum behavior and dry matter intake identify dairy cows at risk for metritis. *J Dairy Sci.* 2007; 90:3220-33.
51. Hayirli A, Grummer RR, Nordheim EV, et al. Animal and dietary factors affecting feed intake during the prefresh transition period in Holsteins. *J Dairy Sci.* 2002;85:3430-43.
52. Fronk T J, Schultz LH, Hardie AR. Effect of dry period overconditioning on subsequent metabolic disorders and performance of dairy cows. *J. Dairy Sci.* 1980; 63:1080.
53. Treacher RJ, Reid IM, Roberts CJ. Effect of body condition at calving on the health and performance of dairy cows. *Anim. Prod.* 1986; 43:1-6.
54. Ribeiro ES, Lima FS, Greco LF, et al. Prevalence of periparturient diseases and effects on fertility of seasonally calving grazing dairy cows supplemented with concentrates. *J. Dairy Sci.* 2013; 96:5682-97.
55. Hansen LB, Freeman AE, Berger PJ. Yield and fertility relationships in dairy cattle. *J. Dairy Sci.* 1983; 66:293-305.

56. Rauw WM, Kanis E, Noordhuizen-Stassen EN, et al. Undesirable side effects of selection for high production efficiency in farm animals: A review. *Livest. Prod. Sci.* 1998; 56:15-33.
57. Hansen LB. Consequences of selection for milk yield from a geneticists viewpoint. *J. Dairy Sci.* 2000; 83: 1145-50.
58. Jones WP, Hansen LB, Chester-Jones H. Response of health care to selection for milk yield of dairy cattle. *J. Dairy Sci.* 1994; 77:3137-52.
59. Crooker BA, Weber WJ, Ma LS, et al. Effect of energy balance and selection for milk yield on the somatotrophic axis of the lactating Holstein cow: Endocrine profiles and hepatic gene expression. In: *Energy Metabolism of Animals (15th Symposium)*. 2001. EAAP Energy Symposium, Snekkersten, Denmark. EAAP Publ. No. 103. Pp 345-348.
60. Lucy MC, Crooker BA. Physiological and genetic differences between low and high index dairy cows. In: *Fertility in the High Producing Dairy Cow*. 2001. British Society of Animal Science Occas. Publ. No. 26. Vol 1. Galway, Ireland. Pp 223-236.
61. Weber WJ, Wallace CR, Hansen LB, et al. Effects of genetic selection for milk yield on somatotropin, insulin-like growth factor-I, and placental lactogen in Holstein cows. *J Dairy Sci.* 2007; 90:3314-25.
62. Butler WR. Nutritional interactions with reproductive performance in dairy cattle. *Anim. Reprod. Sci.* 2000; 60-61:449-57.
63. Weber, W. J., S. J. Kolath, M. C. Lucy, H. Chester-Jones, L. B. Hansen and B. A. Crooker. 2003. Effect of genetic potential for milk yield on the onset of reproductive activity and corpus luteum function in Holstein cows. *J. Dairy Sci.* 86(Suppl. 1):238(Abstr.)

64. Sangsritavong S, Combs DK, Sartori R, et al. High feed intake increases liver blood flow and metabolism of progesterone and estradiol-17beta in dairy cattle. *J Dairy Sci.* 2002; 85:2831-42.
65. Bello NM, Stevenson JS, Tempelman RJ. Invited review: milk production and reproductive performance: modern interdisciplinary insights into an enduring axiom. *J. Dairy Sci.* 2012; 95:5461-75.
66. Berry DP, Wall E, Pryce JE. Genetics and genomics of reproductive performance in dairy and beef cattle. *Animal*, 2014; 8 (Suppl 1):105–21.
67. Parker Gaddis KL, Cole JB, Clay JS. Genomic selection for producer-recorded health event data in US dairy cattle. *J. Dairy Sci.* 2014; 97:3190-9.
68. Carthy, TR, Ryan DP, Fitzgerald AM, et al. Genetic parameters of ovarian and uterine reproductive traits in dairy cows. *J. Dairy Sci.* 2014; 98:4095–106.
69. Bamber RL, Shook GE, Wiltbank MC, et al. Genetic parameters for anovulation and pregnancy loss in dairy cattle. *J. Dairy Sci.* 2009, 92:5739-53.
70. Darwash AO, Lamming GE, Woolliams JA. Estimation of genetic variation in the interval from calving to postpartum ovulation of dairy cows. *J. Dairy Sci.* 1997; 80:1227-34.
71. Schnabel RD, Sonstegard TS, Taylor JF. Whole genome scan to detect QTL for milk production, conformation and functional traits in two U.S. Holstein families. *Anim. Genet.* 2005; 36:408-16.
72. Weigel KA, Van Tassell CP, O'Connell JR, et al. Prediction of unobserved single nucleotide polymorphism genotypes of Jersey cattle using reference panes and population-based imputation algorithms. *J. Dairy Sci.* 2010; 93: 2229-38.

73. Dasonneville R, Brondum RF, Druet T, et al. Effect of imputing markers from a low-density chip on the reliability of genomic breeding values in Holstein populations. *J. Dairy Sci.* 2011; 94: 3679-86.
74. Schaeffer LR. Strategy for applying genome-wide selection in dairy cattle. *J. Anim. Breed. Genet.* 2006; 123:218-23.
75. Olson KM, VanRaden PM, Tooker ME. Multibreed genomic evaluations using purebred Holsteins, Jerseys, and Brown Swiss. *J Dairy Sci.* 2012; 95:5378-83.
76. Weigel KA, Hoffman PC, Herring W, et al. Potential gains in lifetime net merit from genomic testing of cows, heifers, and calves on commercial dairy farms. *J. Dairy Sci.* 2012; 95: 2215
77. Claus MPL, Bijma P, Veerkamp RF. Evaluation of genomic selection for replacement strategies using selection index theory. *J. Dairy Sci.* 2015; 98: 6499-509.

Table 1. Association between body condition score change during the dry period and incidence of postpartum health disorders, removal from the herd within 30 and 60 DIM, and rate of removal from the herd up to 305 d postpartum.

Items	Body condition score change during the dry period				<i>P</i> – value
	$\geq +0.25$	0	-0.25 a -0.5	≤ -0.75	
Cows, n (%)	1,384 (15%)	3,852 (43%)	3,551 (40%)	202 (2%)	
RFM, %	3.0 ^{a,A}	3.4 ^{a,A}	5.5 ^b	7.4 ^B	< 0.01
Metritis, %	10.0 ^a	12.3 ^a	15.7 ^{b,A}	20.8 ^{b,B}	< 0.01
GI disorders, %	2.6 ^a	2.4 ^a	4.8 ^b	9.9 ^c	< 0.01
At least one disease, %	15.0 ^{a,c,A}	17.4 ^a	24.7 ^{b,B}	34.2 ^c	< 0.01
Removal by 30 DIM, %	3.0 ^a	2.8 ^a	5.1 ^b	12.9 ^c	< 0.01
Removal by 60 DIM, %	5.1 ^a	4.5 ^a	7.6 ^b	15.4 ^c	< 0.01
Culling rate	0.88 ^a	Reference ^a	1.14 ^b	1.73 ^c	< 0.01
(AHR and 95% CI)	(0.77, 1.01)		(1.04, 1.25)	(1.36, 2.19)	

Figure Legends

Figure 1a. Average calving interval in Holstein and Jersey cows in Dairy Herd Improvement programs in United States. Data from Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013 (<https://www.cdcb.us/publish/dhi/current/reproall.html>).

Figure 1b. Average pregnancy per artificial insemination (AI; solid line) and interval from calving to first postpartum AI (dashed line) among Holstein cows enrolled in Dairy Herd Improvement programs in United States. Data from Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013 (<https://www.cdcb.us/publish/dhi/current/reproall.html>).

Figure 1c. Figure 1C. Average pregnancy per artificial insemination (AI; solid line) and interval from calving to first postpartum AI (dashed line) among Jersey cows enrolled in Dairy Herd Improvement programs in United States. Data from Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013 (<https://www.cdcb.us/publish/dhi/current/reproall.html>).

Figure 1d. Average daughter pregnancy rate of Holstein and Jersey sires. Data from Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013 (<https://www.cdcb.us/publish/dhi/current/reproall.html>).

Figure 2. Milk yield (kg/day; □), dry matter intake (DMI, kg/day; ○), and milk income over feed cost (IOFC, \$of income/\$of feed consumed; ●) according to days in milk. Milk price was set at 0.35/kg in all scenarios. Panel A: high-producing herd (12,500 kg/305 days of lactation) with feed price at \$0.35/kg of dry matter. Panel B: depicts a moderate-producing herd (9,000 kg/305 days of lactation) and feed price at \$0.29/kg of dry matter. From Ribeiro ES, Galvão KN, Thatcher WW, et al. Economic aspects of applying reproductive technologies to dairy herds. Anim Reprod 2012; 9:370-87, with permission.

Figure 3. Lactation curve and resulting milk production per day of calving interval (CI) according to days in milk at pregnancy in a high-producing herd (12,500 kg in 305 days of lactation - Panel A) and in a moderate-producing herd (9,000 kg of milk in 305 days of lactation - Panel B). From Ribeiro ES, Galvão KN, Thatcher WW, et al. Economic aspects of applying reproductive technologies to dairy herds. Anim Reprod 2012; 9:370-87, with permission.

Figure 4. Observed voluntary waiting period in dairy farms in the United States according to a recent survey. Data from Norman HD, Walton LM, Durr J. Reproductive status of cows in Dairy Herd Improvement programs and bred using artificial insemination. 2013 (<https://www.cdcb.us/publish/dhi/current/reproall.html>).

Figure 5. Estimated voluntary waiting period (VWP) to achieve median days open (DOPN) of 110 or 130 days according to the 21-d pregnancy rate (breeding cycles kept constant = 13). Negative VWP values are the result of calculations and not possible to achieve.

Figure 6. Profits per cow per year (\$/cow/year) of cows subjected to one of ten breeding programs. Bars represent the profit per cow per year according to each reproductive program calculated using milk price at \$0.44/kg. Dashed lines represent median days open. Data from Galvão KN, Federico P, De Vries A, Schuenemann GM. Economic comparison of reproductive programs for dairy herds using estrus detection, timed artificial insemination, or a combination. J. Dairy Sci. 2013; 96: 2681-93.

Figure 7. Proportion of variation in BCS at dry-off accounted for by 305-d mature equivalent milk yield, interval from calving to pregnancy (days open), lactation, and season (winter vs. summer). R^2 of multivariate model = 0.184.

Figure 8. Proportion of variation in BCS loss accounted for by herd, lactation, season (winter vs. summer), length of the dry period, BCS at dry-off, calf sex, and number of calves (singleton vs. twins). R^2 of multivariate model = 0.387%

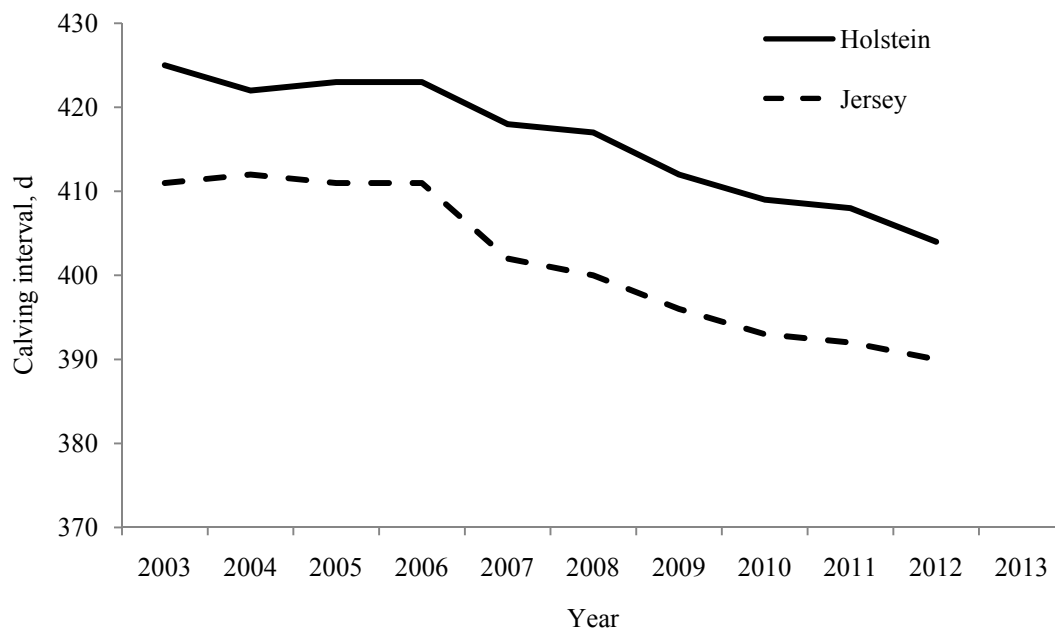


Figure 1A. Average calving interval in Holstein and Jersey cows in Dairy Herd Improvement programs in United States¹⁴.

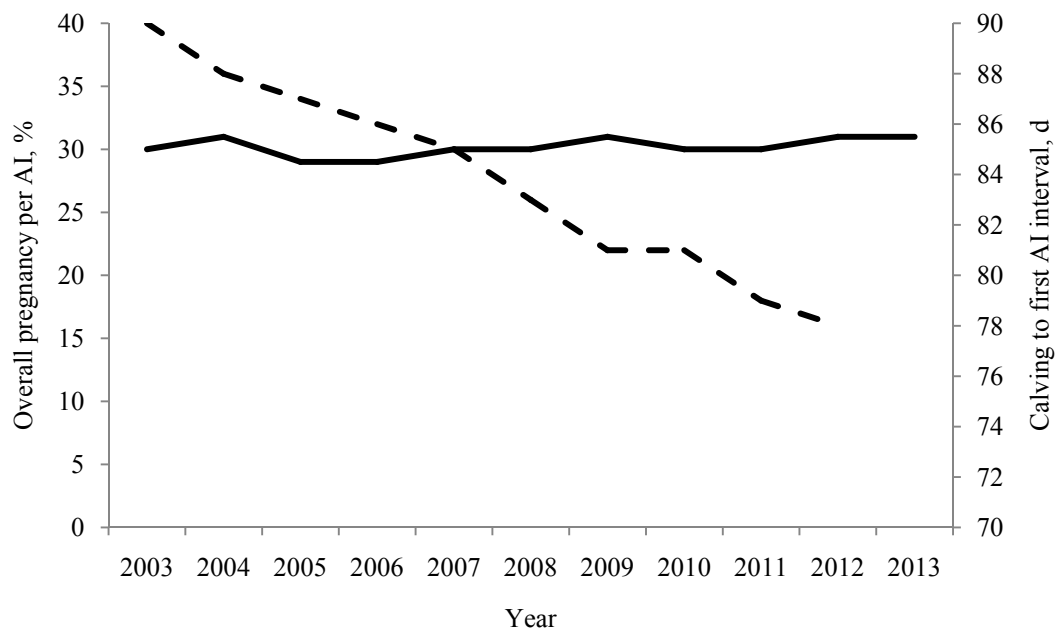


Figure 1B. Average pregnancy per artificial insemination (AI; solid line) and interval from calving to first postpartum AI (dashed line) among Holstein cows enrolled in Dairy Herd Improvement programs in United States¹⁴.

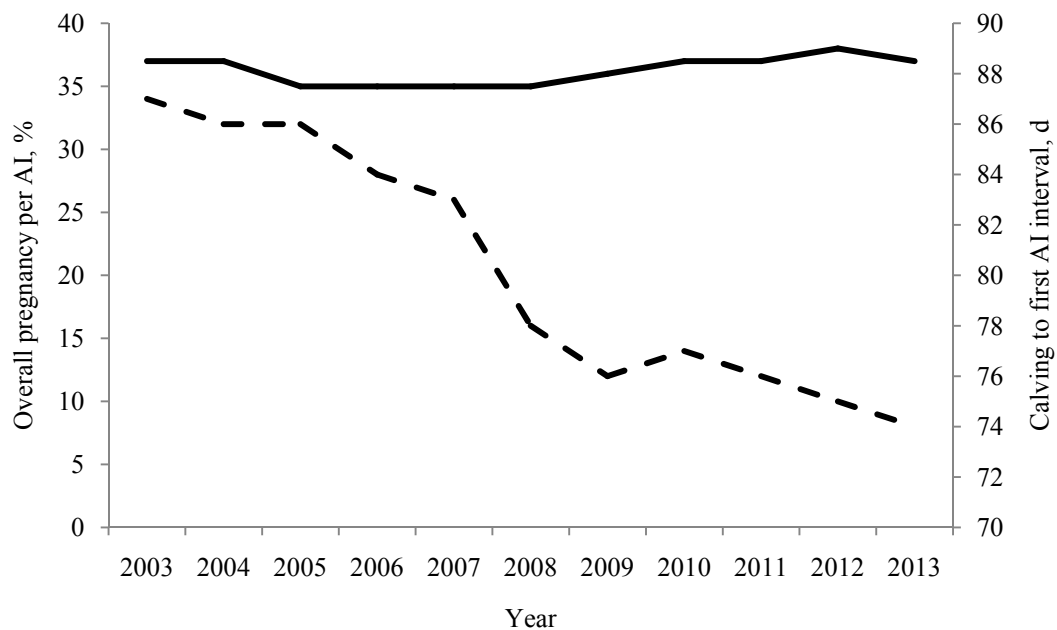


Figure 1C. Average pregnancy per artificial insemination (AI; solid line) and interval from calving to first postpartum AI (dashed line) among Jersey cows enrolled in Dairy Herd Improvement programs in United States¹⁴.

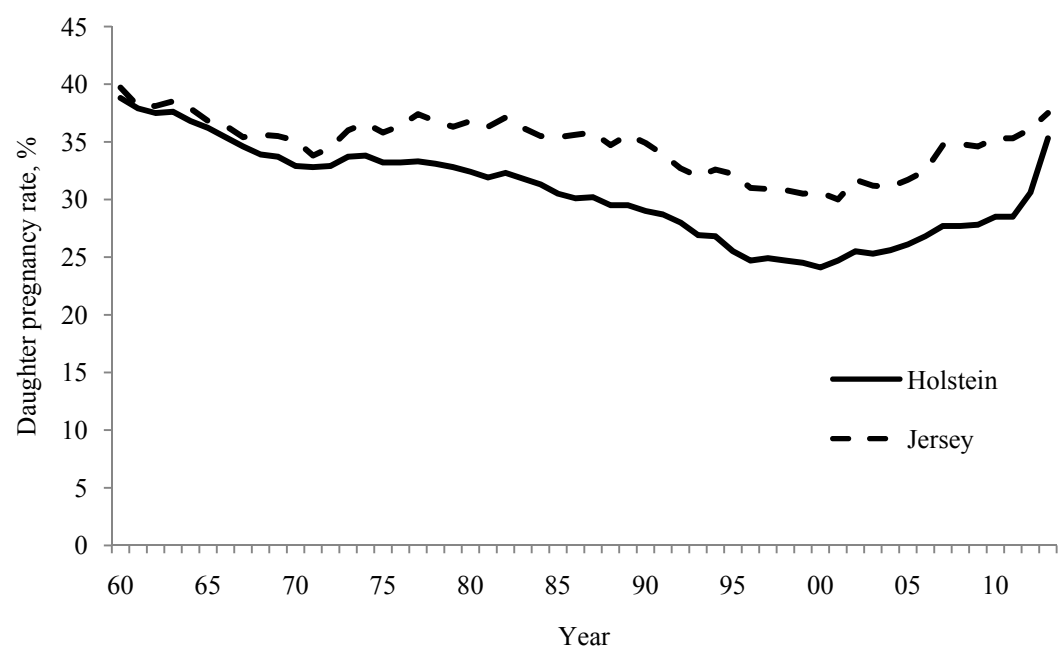


Figure 1D. Average daughter pregnancy rate of Holstein and Jersey sires¹⁴.

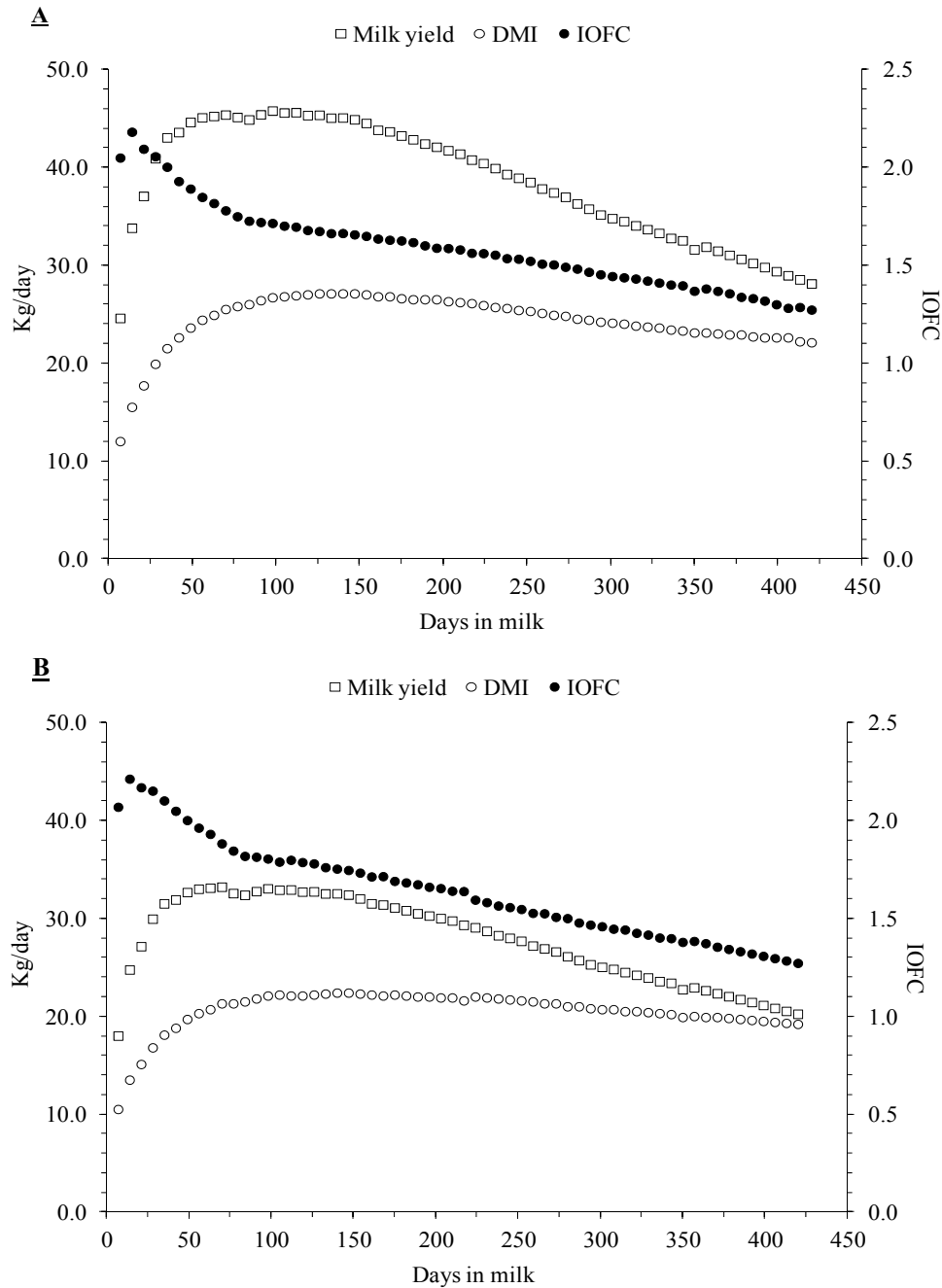


Figure 2. Milk yield (kg/day; □), dry matter intake (DMI, kg/day; ○), and milk income over feed cost (IOFC, \$of income/\$of feed consumed; •) according to days in milk. Milk price was set at 0.35/kg in all scenarios. Panel A: high-producing herd (12,500 kg/305 days of lactation) with feed price at \$0.35/kg of dry matter. Panel B: depicts a moderate-producing herd (9,000 kg/305 days of lactation) and feed price at \$0.29/kg of dry matter¹⁶.

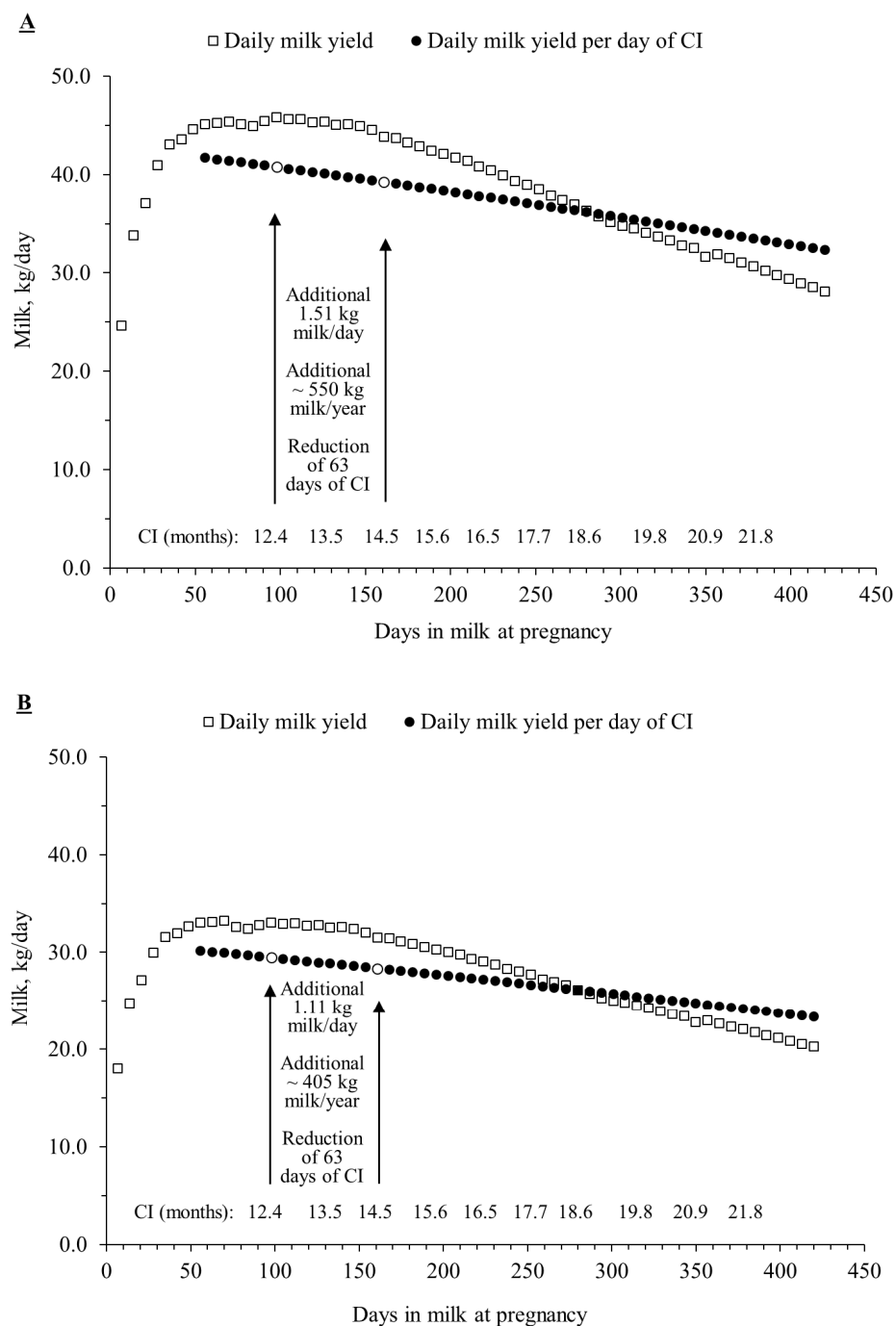


Figure 3. Lactation curve and resulting milk production per day of calving interval (CI) according to days in milk at pregnancy in a high-producing herd (12,500 kg in 305 days of lactation - Panel A) and in a moderate-producing herd (9,000 kg of milk in 305 days of lactation - Panel B)¹⁶.

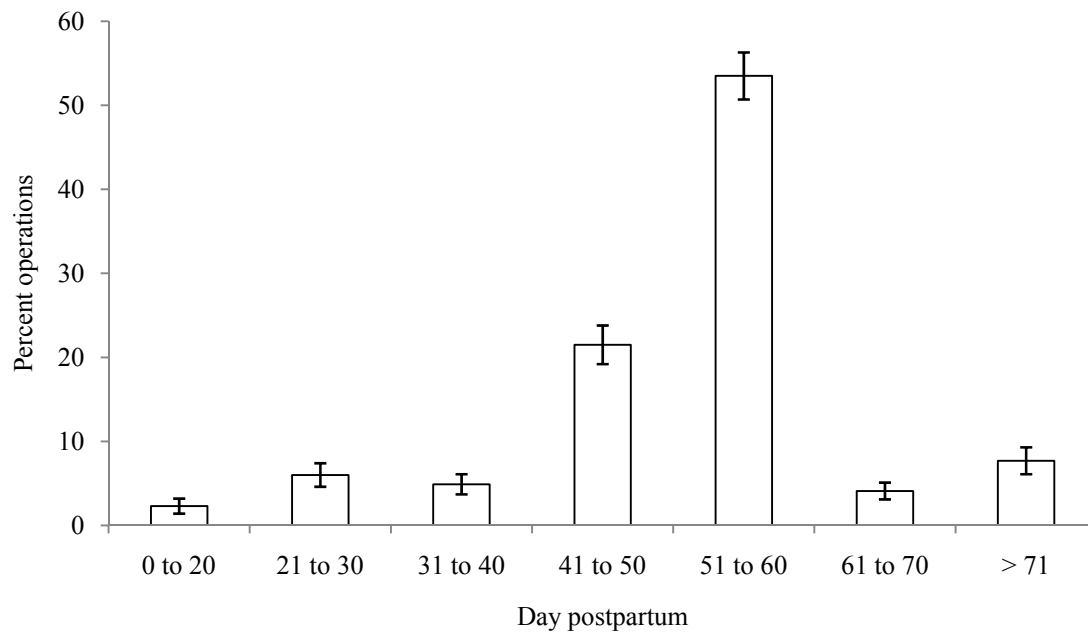


Figure 4. Observed voluntary waiting period in dairy farms in the United States according to a recent survey¹⁴.

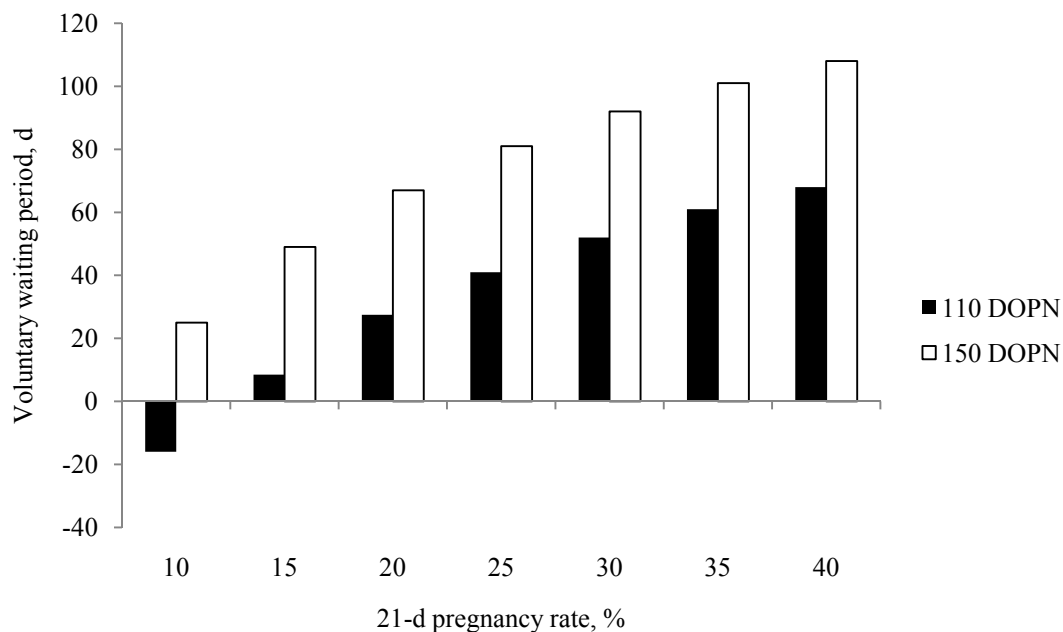


Figure 5. Estimated voluntary waiting period (VWP) to achieve median days open (DOPN) of 110 or 130 days according to the 21-d pregnancy rate (breeding cycles kept constant = 13). Negative VWP values are the result of calculations and not possible to achieve.

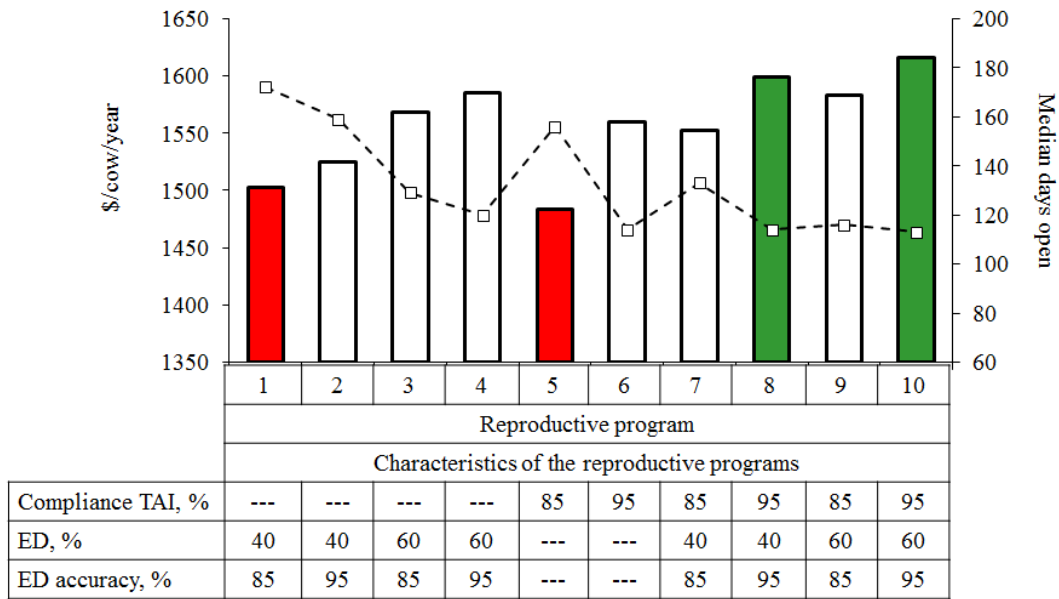


Figure 6. Profits per cow per year (\$/cow/year) of cows subjected to one of ten breeding programs. Bars represent the profit per cow per year according to each reproductive program calculated using milk price at \$0.44/kg. Dashed lines represent median days open⁴.

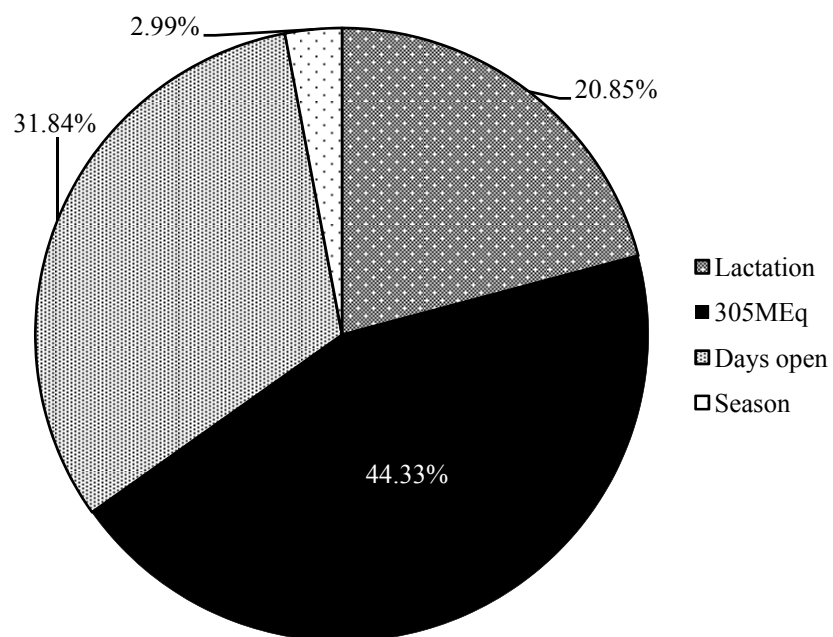


Figure 7. Proportion of variation in BCS at dry-off accounted for by 305-d mature equivalent milk yield, interval from calving to pregnancy (days open), lactation, and season (winter vs. summer). R^2 of multivariate model = 0.184.

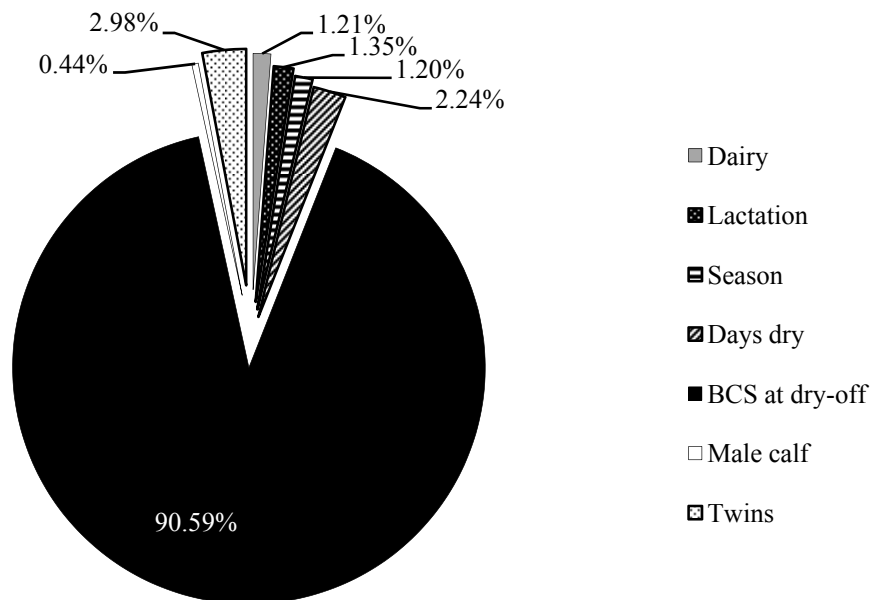


Figure 8. Proportion of variation in BCS loss accounted for by herd, lactation, season (winter vs. summer), length of the dry period, BCS at dry-off, calf sex, and number of calves (singleton vs. twins). R^2 of multivariate model = 0.387.